Appendix O

Small research projects

In this Appendix, we list a few small research projects. These projects involve the development of your own program. A possible strategy is to use the source code of one of the Case Studies as the starting point. It is our experience that the completion of such a research project takes about two weeks, depending on your experience.

O.1 Adsorption in porous media

In this project, we will investigate the adsorption behavior in porous media. As a model we use a slit-like pore. The interactions with the pore are given by

$$U(z) = \begin{cases} 0 & 0 < z < L \\ \infty & \text{otherwise} \end{cases}, \tag{O.1.1}$$

where L is the width of the slit. We will investigate the adsorption of methane, which we model with a Lennard-Jones potential. The starting point is Case Study 9, which uses a program to simulate the Lennard-Jones fluid in the grand-canonical ensemble. The project is to develop a program to simulate an adsorption isotherm of methane in the slit-like pore.

Before starting to program you may want to think about the following points:

1. What is the geometry of the system and how should one apply the periodic boundary conditions?
2. What is the type of Lennard-Jones potential (truncated, truncated and shifted, with or without tail corrections)? And what are the parameters for modeling methane?
3. How is an adsorption isotherm defined thermodynamically?

With your program, try to answer the following questions:

1. Compute the density profile (density as function on the distance z between the plates) for $L = 1, 2, 5$, and 10 for $\rho = 0.7$ and $T = 2.0$. The total number of particles should be of the order of 100 to 500 for the widest slit.
2. Compute the excess chemical potential and chemical potential as a function of the distance between the plates for $L = 1, 2, 5$, and 10 for $\rho = 0.6$ and $T = 2.0$. Try to explain the differences.
3. Compute the adsorption isotherms for $L = 2$ and 5 for $T = 2.0$ and 0.8. Try to explain the results.

O.2 Transport properties of liquids

Molecular Dynamics simulations can be used to compute the transport properties of liquids. Examples of these transport properties are the (self-) diffusion coefficient and the viscosity. In Case Study 5 the results for the diffusion coefficient are shown. In this project we extend these results to mixtures of Lennard-Jones (LJ) molecules. Before starting to program you may want to think about the following points:

1. The generalization of the diffusion coefficient from a pure component to a mixture is not trivial. Try to find in the literature how one should define a diffusion coefficient of a mixture and how can one compute this in a simulation. See, for example, refs. [59,784].
2. What is the type of LJ potential (truncated, truncated and shifted, with or without tail corrections)? And what are the parameters for modeling argon and krypton?

   With your program, try to answer the following questions:

1. Compute the pressure, viscosity, and diffusion coefficient of the LJ fluid at $T = 1.0$, 1.5, and 2.0 for $\rho = 0.7$.
2. Compute the diffusion coefficients D_{11}, D_{12}, and D_{22} for a mixture of 50%-50% LJ particles in which the components 1 and 2 have the same interactions ($\sigma_{12} = \sigma_{11} = \sigma_{22}$ and $\epsilon_{12} = \epsilon_{11} = \epsilon_{12}$) but carry a different color. This means that the particles are labeled. Experimentally this could be done by radioactive labeling.
3. Compute the pressure, viscosity, and diffusion coefficient of the LJ fluid at $T = 1.0, 1.5$, and 2.0 for $\rho = 0.7$. Compute the diffusion coefficients D_{11}, D_{12}, and D_{22} for a mixture of 50-50% LJ particles, but now for a system of 50% argon and 50% krypton (use the parameters of argon to compute the reduced temperatures). Does the Einstein equation for the diffusivity hold for this system?

O.3 Diffusion in a porous medium

The behavior of liquid in confined geometries is different from the behavior in the bulk liquid. In the project we investigate the diffusion coefficient in a cylindrical pore. The starting point of the present study is Case Study 5, which describes the simulations of the dynamic properties of a Lennard-Jones fluid. The interactions with the walls are described with a repulsive potential:

$$U(r) = \begin{cases} \epsilon \left(\frac{\sigma}{L-r} \right)^{10} & 0 \le r \le L \\ \infty & r > L \end{cases}, \tag{O.3.1}$$

where L is a radius characterizing the size of the pore and the center of the cylinder is located at $r = 0$ and $\epsilon > 0$, $\sigma > 0$. Some questions that one should answer before one starts programming are the following:

1. Is the potential for the interactions with the walls appropriate for a Molecular Dynamics simulation?
2. What is the volume of the pore as a function of the parameter L?
3. What is the dimension of our problem? Do we have diffusion in 1, 2, or 3 dimensions?

In the first part of the project we study the diffusion in a smooth pore as defined by the above potential as a function of the pore diameter.

1. Compute the diffusion coefficient of a bulk Lennard-Jones liquid for $\rho = 0.6$ and $T = 2.0$ and 1.5. Since the program uses an NVE ensemble, it is not possible to simulate at exactly the requested temperature. However, one can ensure to be close to this temperature by an appropriate equilibration of the system (this is also the case for the following two questions) during the first part of the MD simulation.
2. Compute the density as a function of the distance from the center of the pore for $\rho = 0.6$ and $T = 2.0$ and 1.5 and $L = 5$ and 2. Interpret the results.
3. Compute the diffusion coefficient for $\rho = 0.6$ and $T = 2.0$ and $T = 1.5$ and $L = 5$, 2, and 1. Interpret the results. The interpretation is not trivial.
4. The above calculations have been performed using the NVE ensemble. This implies that there is no coupling with the atoms of the walls. In a real system the walls are not smooth and can exchange heat with the adsorbed molecules. A possible way of modeling this is to assume that we have an Andersen thermostat in the boundary layer with the wall. Investigate how the results depend on the thickness of the boundary layer and the constant ν of the Andersen algorithm.

The next step is to model the corrugation caused by the atoms. This corrugation could be a term:

$$U(z, r) = A \sin^2(\pi z/\sigma_w) \exp\left[-\left(\frac{r - L}{L_0}\right)^2\right], \tag{O.3.2}$$

where z is the distance to the wall, and σ_w is a term characterizing the size of the atoms of the wall and A is the strength of the interaction. The exponential is added to ensure that the potential is localized close to the walls of the cylinder. Investigate the diffusion coefficient as a function of the parameters σ_w and A both in the NVE and in the Andersen thermostat cases.

O.4 Multiple-time-step integrators

The time step in a Molecular Dynamics simulation strongly depends on the steepness of the potential energy surface. However, most potentials like the Lennard-Jones potential are steep at short distances. As short-range interactions can be computed very fast, it would be interesting to use a multiple-time-step integration algorithm, in which short-range (computationally cheap) interactions are computed every time step and in which long-range (computationally expensive) interactions are evaluated every n time steps ($n > 1$). Recently, there has been a considerable effort to construct time-reversible multiple-time-step algorithms [117,126].

1. Why is it important to use time-reversible integration schemes in MD?
2. Modify Case Study 4 in such a way that pairwise interactions are calculated using a Verlet neighbor list. For every particle, a list is made of neighboring particles within a distance of $r_{\text{cut}} + \Delta$. All lists have to be updated only when the displacement of a single particle is larger than $\Delta/2$. Hint: The algorithm[1] in book of Allen and Tildesley [21] is a good starting point.
3. Investigate how the CPU time per time step depends on the size of Δ for various system sizes. Compare your results with Table 5.1 from ref. [21].
4. Modify the code in such a way that the NVE multiple-time-step algorithm of ref. [126] is used to integrate the equations of motion. You will have to use separate neighbor lists for the short-range and the long-range parts of the potential.
5. Why does one have to use a switching function in this algorithm? Why is it a good idea to use a linear interpolation scheme to compute the switching function from ref. [126]?
6. Make a detailed comparison between this algorithm and the standard Leap-Frog integrator (with the use of a neighbor list) at the same energy drift.

O.5 Thermodynamic integration

The difference in free energy between state A and state B can be calculated by thermodynamic integration:

$$F_A - F_B = \int_{\lambda=0}^{\lambda=1} d\lambda \left\langle \frac{\partial U(\lambda)}{\partial \lambda} \right\rangle_\lambda , \tag{O.5.1}$$

in which $\lambda = 1$ in state A and $\lambda = 0$ in state B. In order to calculate the excess chemical potential of a Lennard-Jones system, we might use the following modified potential[2] [785]:

$$U(r, \lambda) = 4\epsilon \left(\lambda^5 \left(\frac{\sigma}{r}\right)^{12} - \lambda^3 \left(\frac{\sigma}{r}\right)^6 \right). \tag{O.5.2}$$

Recall that the excess chemical potential is the difference in chemical potential between a real gas ($\lambda = 1$) and an ideal gas ($\lambda = 0$).

1. Make a plot of the modified LJ potential for various values of λ.

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[1] https://github.com/Allen-Tildesley/examples/blob/master/md_lj_vl_module.f90.

[2] As the reader can easily verify, this modified potential is still a normal LJ 12-6 potential, but with different values of ϵ and σ.

2. Show that

$$\left(\frac{\partial^2 F}{\partial \lambda^2}\right) < 0 \tag{O.5.3}$$

when

$$U = U_0 \lambda + U_1 (1 - \lambda) . \tag{O.5.4}$$

3. Derive Eq. (O.5.1).
4. Modify the code of Case Study 1 for this modified potential.
5. Perform the thermodynamic integration and compare your results with the conventional particle insertion method.
6. Calculate the chemical potential as a function of the density by scaling σ.